1 Action of the QFT on basis states

The action of the QFT on an N-dimensional Hilbert space ($N = 2^n$) with computational basis states $|j\rangle$ is defined through the relation

$$\mathrm{QFT}_N |j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i jk/N} |k\rangle .$$

(a) Show that for $n = 1$ ($N = 2$), the matrix QFT_2 is identical to the Hadamard gate H.

We will denote by Q_{kj} the matrix element

$$Q_{kj} = \langle k|\mathrm{QFT}_2|j\rangle = \frac{1}{\sqrt{2}} e^{\pi i jk} = \frac{1}{\sqrt{2}} (-1)^{jk}.$$

For $k = 0, 1$ and $j = 0, 1$ (relevant to the single-qubit case), we then have $Q_{00} = Q_{01} = Q_{10} = 1/\sqrt{2}$ and $Q_{11} = -1/\sqrt{2}$. In the basis $\{|0\rangle, |1\rangle\}$, the matrix QFT_2 is then given by

$$\mathrm{QFT}_2 = \begin{bmatrix} Q_{00} & Q_{01} \\ Q_{10} & Q_{11} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix},$$

corresponding also to the matrix of a Hadamard.

(b) Write out the matrix QFT_4 in the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. Write your matrix in terms of powers of $\omega = e^{2\pi i/4} = i$.

In this case, we have matrix elements given by

$$Q_{kj} = \frac{1}{2} e^{\frac{\pi}{2} ijk} = \frac{1}{2} i^{jk},$$

where $j, k \in \{0, 1, 2, 3\}$. In the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, we then have

$$\begin{aligned} \mathrm{QFT}_4 &= \begin{bmatrix} Q_{00} & Q_{01} & Q_{02} & Q_{03} \\ Q_{10} & Q_{11} & Q_{12} & Q_{13} \\ Q_{20} & Q_{21} & Q_{22} & Q_{23} \\ Q_{30} & Q_{31} & Q_{32} & Q_{33} \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix}. \end{aligned}$$

(c) Evaluate $\mathrm{QFT}_8 |000\rangle$ ($j = 0$). Is the resulting state a product state?

With $j = 0$, we get a uniform superposition over all computational basis states,

$$\mathrm{QFT}_8 |0\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} |k\rangle = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right).$$

As shown above, this superposition can also be written as the product state $|+\rangle^{\otimes 3}$.

2 Circuit representation and complexity

A QFT circuit can be expressed in terms of Hadamard gates and controlled rotation gates R_j, where

$$R_j = \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^j} \end{pmatrix}.$$

(a) Give the number of Hadamards and controlled-R_j gates needed to implement a three-qubit QFT circuit. How about four qubits? To keep things simple, disregard the SWAP gates used to invert the bit ordering.

For a three-qubit QFT circuit, we need three Hadamards. The first qubit is the target qubit in two two-qubit gates, while the second qubit is the target of one two-qubit gate. This gives a total of three two-qubit gates. For a four-qubit QFT circuit, we need four Hadamards. In this case, the first qubit is the target for three two-qubit gates, while the second (third) is involved in two (one) two-qubit gate(s). This gives a total of six two-qubit gates.

(b) Prove that an n-qubit QFT circuit requires $O(n^2)$ elementary gate operations.

For an n-qubit QFT circuit, the number of Hadamards required is n. Each qubit Q_i with $i \in \{1, \ldots, n\}$ will also serve as the target qubit for $n - i$ two-qubit gates. The total number N_{gates} of gates is then

$$\begin{aligned} N_{\text{gates}} &= n + \sum_{i=1}^{n} (n - i) \\ &= n + n^2 - \sum_{i=1}^{n} i \\ &= n + n^2 - \frac{n(n+1)}{2} \\ &= \frac{n^2 + n}{2}. \end{aligned}$$

The leading scaling is therefore $N_{\text{gates}} = O(n^2)$.

3 Periodicity and Interference

Consider a three-qubit system ($N = 8$) with computational basis states $|j\rangle$ labelled by $j = 0, \ldots, 7$. A three-qubit state is said to have period r if its nonzero amplitudes occur at values separated by r (mod 8). Let's assume the qubits are initialized in the state $|\Psi\rangle$ given by

$$|\Psi\rangle = \frac{1}{\sqrt{2}} \left(|1\rangle + |5\rangle \right).$$

(a) What is the period r of this state?

Since $5 - 1 = 4$ and $5 + 4 \equiv 1 \pmod 8$, this state has period $r = 4$.

(b) Compute $\text{QFT}_8 |\Psi\rangle$. You may find it easier to evaluate the transform on $|1\rangle$ and $|5\rangle$ separately.

First, we write the action of QFT_8 on the two basis states separately, giving

$$\mathrm{QFT}_8 |1\rangle = \frac{1}{\sqrt{8}} \sum_{k=0}^{7} \omega^k |k\rangle ,$$
$$\mathrm{QFT}_8 |5\rangle = \frac{1}{\sqrt{8}} \sum_{k=0}^{7} \omega^{5k} |k\rangle ,$$

where $\omega = e^{i\pi/4}$. Combining these results gives

$$\begin{aligned} \mathrm{QFT}_8 |\Psi\rangle &= \frac{1}{4} \sum_{k=0}^{7} \omega^k (1 + \omega^{4k}) |k\rangle \\ &= \frac{1}{4} \sum_{k=0}^{7} \omega^k (1 + (-1)^k) |k\rangle \\ &= \frac{1}{2} \left(|0\rangle + i |2\rangle - |4\rangle - i |6\rangle \right) . \end{aligned}$$

(c) Which computational basis states appear in $\mathrm{QFT}_8 |\Psi\rangle$? How do their indices relate to N and r?

If k is odd, then $|k\rangle$ does not appear in $\mathrm{QFT}_8 |\Psi\rangle$ due to destructive interference, while states indexed by even k do appear. Note that the period of $\mathrm{QFT}_8 |\Psi\rangle$ is $2 = N/r$.

4 Operator Diagonalization

Let X_N be the cyclic shift operator defined by its action on computational basis states $|j\rangle$ according to

$$X_N |j\rangle = |(j+1) \bmod N\rangle , \quad j = 0, 1, \ldots, N-1.$$

Furthermore, let Z_N be the phase operator defined by

$$Z_N = \sum_{k=0}^{N-1} e^{2\pi i k/N} |k\rangle\langle k| .$$

Show that the quantum Fourier transform diagonalizes X_N. In particular, show that

$$\mathrm{QFT}_N X_N \mathrm{QFT}_N^\dagger = Z_N .$$

We start by considering the action of $\mathrm{QFT}_N^\dagger$ on the computational basis state $|k\rangle$, given by

$$\mathrm{QFT}_N^\dagger |k\rangle = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{-2\pi i j k/N} |j\rangle .$$

Using the definition given in the problem statement, applying the shift operator X_N to this state gives

$$X_N \left(\mathrm{QFT}_N^\dagger |k\rangle \right) = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{-2\pi i j k/N} |(j+1) \bmod N\rangle .$$

We now switch the index of summation to $j' = (j+1) \bmod N$. Since j ranges from 0 to $N-1$, j' also ranges from 0 to $N-1$. Applied to the equation above, this change of variables gives

$$\begin{aligned} X_N \left(\mathrm{QFT}_N^\dagger |k\rangle \right) &= \frac{1}{\sqrt{N}} \sum_{j'=0}^{N-1} e^{2\pi i k/N} e^{-2\pi i j' k/N} |j'\rangle \\ &= e^{2\pi i k/N} \mathrm{QFT}_N^\dagger |k\rangle \,, \end{aligned}$$

where we have used the fact that $j = j' - 1 + nN$ for some integer $n \in \mathbb{Z}$. Acting with QFT_N from the left, we then have

$$\begin{aligned} \mathrm{QFT}_N X_N \mathrm{QFT}_N^\dagger |k\rangle &= e^{2\pi i k/N} |k\rangle \\ &= Z_N |k\rangle \,. \end{aligned}$$

Since this relation holds for every basis state $|k\rangle$, we then arrive at the operator identity

$$\mathrm{QFT}_N X_N \mathrm{QFT}_N^\dagger = Z_N.$$